

Guaranteed-Companion Seating: Current Implementation and Evidence Guide

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This report specifies the visual-simulation implementation after the revisions of 11 and 13 September 2026. The formal manuscript, including proofs, related work, figures, experimental results, and limitations, is `article/article.tex`. Historical proposal documents describing maximization of two-friend coverage are superseded, and the study of 8 September 2026 is retained only as historical evidence of the previous implementation. The original external prototype and historical collusion experiment files are unavailable; this repository implements and tests the stated mechanism independently.

1 Inputs, submission states, and validity

Each participant has a directed submitted list. In the defended reference case, a list has at least four distinct non-self names, at least two from the participant's grade, and at most eight names, or is empty. A nonempty list that fails the rule is stored as *pending review*; it remains a visible request and blocks scheduling until an administrator approves it as an exception (the actual list keeps its obligation), asks for a resubmission, or records a withdrawal. Simulation policies (`approve`, `withdraw`) apply one decision to every pending request and record that a policy, not a person, decided. Nothing pads a list. The four states are accepted, pending review, approved exception, and voluntary nonsubmission; participation and admission are reported separately from guarantee satisfaction.

For each rotation the active roster is the set of present participants. Every present participant appears exactly once, tables meet their occupancy targets within physical capacities, grade restrictions hold, staff-prohibited pairs never share a table, placement restrictions hold, and every present participant with an unwaived obligation shares a table with a present, eligible listed companion. A present obligated participant whose every listed name is absent or ineligible is an explicit conflict that stops scheduling until a recorded decision (a per-rotation waiver, a list change, or a change of attendance). Absences reduce occupancy targets one seat at a time, largest tables first within the admissible grade pool. All of these checks are implemented once, in `sim/constraints.py`, and every released chart, cached fallback, and manual adjustment passes through them.

2 Admission screens and their limits

Structural admission screens use the full submitted roster, filtered separately for each requested eligibility state. They are not repeated on each attendance-restricted graph and can conservatively request review despite an attendance-specific solution. Successor closures of at most 64 vertices are checked for a split into guarantee-closed parts. Targeted candidates additionally include strongly connected cores of sizes 2–12 in the original graph and every graph obtained by deleting a single vertex. Each targeted relaxation restores every original outgoing edge of the core, adds all its outside names, and removes only the outside vertices' obligations. If even this easier model cannot separate core members into two supported parts, every valid original chart keeps the core together.

Table 1: Reference configuration. Complete generator, submission, and solver configurations are retained in every trace.

Parameter	Value
Population	257 (132 grade 11; 125 grade 12)
Tables	17×7 and 23×6
Rotations	16, alternating mixed and same grade
Defended study lists	None or 4–8 names; at least 2 same grade
Reference submission states	257 accepted, 0 pending, 0 approved exceptions, 0 voluntary nonsubmission
Objective weights	extra companion 1, $\lambda = 0.1$, repeats 5/2; integer scale 10
CP-SAT formulation	all history pairs, integer extras (count)
Attendance	absence rate 0; conflict policy stop
Popularity / within-grade weight	$\sigma = 0.6 / 9$
Reciprocity target / realized	0.5 / 0.490
Realized within-grade fraction	0.904
Reference communities	$\mu = 0$, $\omega = 0$
Annealing budget / temperature	300,000 iterations / $25 \rightarrow 0.2$ (integer energy)
CP budget / workers	3.5 deterministic-time units / 8
Pre-feasibility budget	20.0 deterministic-time units per state
Runtime versions	Python 3.14.7, NumPy 2.5.2, OR-Tools 9.15.6755

On the first requested roster of each eligibility state, high-risk candidates (unresolved, proved coercive, or exposed only after deleting an anchor) also receive a *full-model separation test*: the complete assignment model of the rotation state (occupancy targets, eligibility, attendance-restricted lists, every obligation, staff constraints) plus the constraint $\sum_{i \in C} x_{it} \leq |C| - 1$ for every table t . A FEASIBLE or OPTIMAL result yields a chart that is validated independently and really seats the group at two or more tables; INFEASIBLE certifies that no valid chart under the current constraints separates the group; UNKNOWN is unresolved. A feasible base witness makes a forcing verdict nonvacuous. If the base status is UNKNOWN, infeasibility of the augmented model excludes separating charts but leaves base feasibility unresolved. Budgets (candidate caps, deterministic time per test, number of full-model tests per state, largest group size) are configurable and every report records how many candidates were discovered, examined, and skipped. A forced group is a certificate about the lists and complete tested constraints, not evidence of intent. Later changed rosters receive feasibility checks, but do not receive fresh full-model coalition-separation searches; those later states remain outside the initial screening coverage.

The demand test and the review gate are unchanged in substance: a self-contained core with n_0 nonsubmitters and n_1 submitters, together with its followers, can occupy at most $n_0 + \lfloor n_1/2 \rfloor$ tables of its admissible grade pool. A screen-enabled year requires review if the input contains pending requests, returns, unresolved checks, rule violations without an approved exception, submitters without eligible names, or full-model forced groups. Recurring co-seated groups across the released charts are reported descriptively for the proposed and the random charts alike.

3 Feasibility certificates and validated fallbacks

Attendance callbacks are materialized once, and only requested eligibility states and actual rosters enter preflight. For each distinct set of hard constraints (the fingerprint covers roster, obligations, eligibility, capacities, occupancy targets, and staff constraints; meeting history is deliberately excluded because it changes scores, not validity) a hard-constraint CP-SAT model is attempted. FEASIBLE or OPTIMAL supplies a witness chart, which is validated independently and cached under the fingerprint; INFEASIBLE proves that the specified model admits no valid chart; UNKNOWN establishes neither. Every released chart is also cached. A cached chart is revalidated against the current inputs before it is used, so a stale entry can never be released.

Cached charts are incumbents for the search: they restart annealing when construction fails to reach feasibility, they seed the CP hint when they score better than annealing, and they are released unchanged when every stage fails or times out. The release status of a rotation is therefore *optimized* (produced by this rotation’s search), *incumbent* (a cached chart that no search result beat), or *fallback* (the search never reached a valid chart of its own). A chart is described as optimal only when CP-SAT returned OPTIMAL under the exact objective. A year ends in exactly one of: scheduled, review required, proven infeasible, or unknown.

4 Objective and solver stages

One specification, `sim/objective.py`, defines the objective for every stage. With s_i the number of present eligible listed companions at the table, ν the number of hard violations, and m_{ab}, α_{ab} the disjoint historical counters,

$$C = w_e \sum_i \max(0, s_i - 1) + 0.1 \sum_{a < b \text{ together}} (5m_{ab} + 2\alpha_{ab}), \quad E = sC + 1000s\nu,$$

where w_e (default 1) is the configurable penalty per additional listed companion and s is the smallest integer making every scaled weight integral ($s = 10$ for the defaults, giving 10 per extra companion, 5 per incidental and 2 per listed prior meeting, and temperature $25 \rightarrow 0.2$). The hard at-least-one requirement is not configurable.

Construction and repair. `pod_greedy` prioritizes constrained participants, forms pods, never pools a prohibited pair, and honours placement restrictions whenever a compatible table has room. `swap_repair` uses guarded swaps and compound moves that reduce the unsatisfied set without orphaning previously satisfied submitters or creating another hard violation. Repair may stop before feasibility.

Annealing. Proposals mix uniform, directed, and group moves. Accepted proposals cannot increase ν ; once feasible, the chain stays feasible. Placement restrictions filter proposals. After an infeasible run of I iterations, up to six repair-and-anneal retries use $\max(I/2, 20000)$ iterations; if the construction path still fails and a validated incumbent exists, annealing restarts from that incumbent.

CP-SAT. Assignment variables have exact-one, occupancy, eligibility, and prohibited-pair constraints. Supporting indicators obey $y_{it} \leq x_{it}$, $y_{it} \leq \sum_{j \in L_i} x_{jt}$, and $\sum_t y_{it} \geq 1$. An integer extras variable $e_i \in [0, |L_i| - 1]$ satisfies $e_i \geq \sum_{j \in L_i} x_{jt} - 1 - |L_i|(1 - x_{it})$ for every table, and a repeat indicator r_{ab} exists for every pair with nonzero history cost, with $r_{ab} \geq x_{at} + x_{bt} - 1$ for every table. CP-SAT minimizes $s w_e \sum_i e_i + \sum_{\{a,b\}} (5m_{ab} + 2\alpha_{ab}) r_{ab}$, whose minimum over the auxiliary variables equals sC for each valid chart at the default repeat weights. A merely feasible auxiliary assignment can overestimate this score. Thus a proven optimum is an optimum of the rotation objective. The hint is the better of the annealed chart and the incumbent. A returned chart is rescored independently; the CP objective can never be below that score and must equal it at OPTIMAL, and a violation of this check is an error. A feasible CP candidate replaces a valid incumbent only if it does not worsen the independent score. The previous-rotation pair surrogate and the two-or-more indicator remain available as explicit options for comparison only. An exploratory summary-only benchmark, without retained returned charts or source fingerprint, reports 257-participant timings (5.0 s mean versus 3.6 s for the historical surrogate on the same incumbents) and worsened the incumbent in 0 of 10 benchmark rotations, whereas the surrogate’s charts were worse under the full objective in 7 rotations.

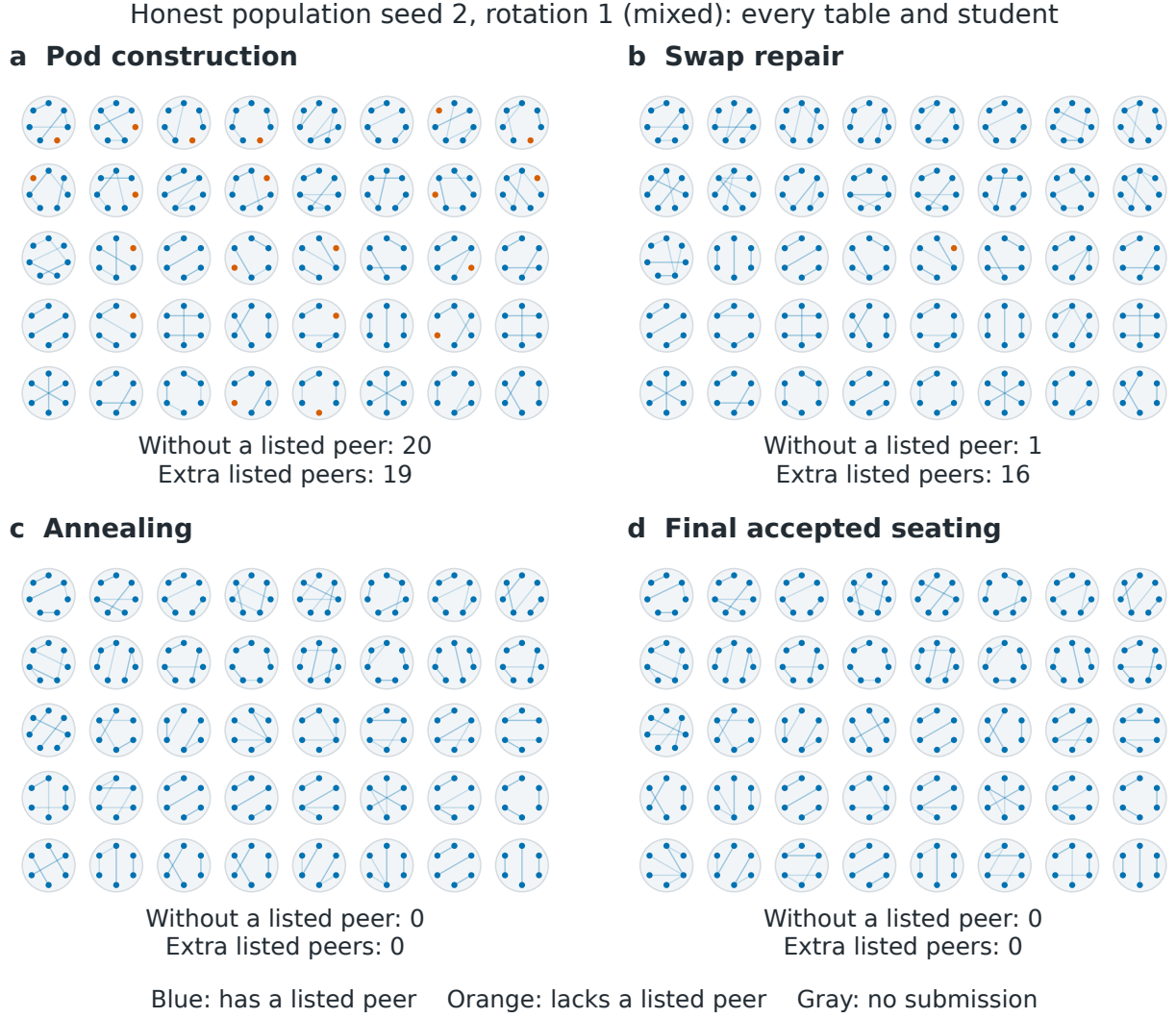


Figure 1: Actual construction, repair, annealing, and final endpoints from one verified rotation. The early stages can violate the companion constraint. These are saved assignments, not fabricated illustrations of a browser optimizer.

Companion-preserving baseline. `-method construct_repair` stops after construction and repair, with the same validated fallbacks and no diversity optimization. It is the simpler comparator in the method-comparison grid.

5 Metrics, baseline, and privacy

Coverage and exactly-one percentages use present participants with unwaived obligations as the denominator. Distinct-peer means use all participants. Per-participant outcomes (distinct peers with minimum and lower percentiles, extra companions, the maximum number of rotations spent with the same listed peer, the share of obligated rotations spent with the most frequent listed peer) are computed for the proposed and for the matched random charts. Coalition member-weighted cluster size is $\sum_G |G \cap C|^2 / |C|$. The random baseline uniformly permutes present participants into the same occupancy targets with the same eligibility states.

The exact privacy probe treats a tablemate set as a hitting constraint only in an observed rotation with a present, unwaived obligation, under a public list cap. A waived rotation imposes no companion constraint. Two further analyses are recorded: the worst logical forcing when only the current chart is distributed, and the co-seating exposure statistic, the fraction of obligated

participants whose most frequent tablemate is a listed peer, for the proposed and random charts. In the reference year the latter is 98.0% against 3.5% under random seating while current-only distribution forces 0 entries. Exposure uses observations across rotations and can be maintained with running counts without retaining full charts. No privacy guarantee is claimed. Student-facing distribution uses `sim/student_export.py`, which contains the assignment of one rotation and nothing private; the draft submission notice and data-handling policy in `docs/policy/` state the limits.

6 Evidence, reproduction, and website

The experiment ledger records a run before executing it. Terminal statuses retain interruptions, input failures, review requirements (including attendance conflicts), unknown outcomes, source changes, and other unsuccessful outcomes. A success includes a full trace, input and configuration identity, effective source-file hashes, full Git identity, dirty status, runtime versions, and a trace checksum. The revised study’s inventory is in `results/revised_experiments.json` and the incomplete method comparison in `results/revised_evaluation.json`; exact commands are in `results/STUDY.md`; the executable source is preserved by effective hash under `results/source_snapshots/`. The evaluation retains 48 attempts and 31 successful traces, with two attempts lacking terminal outcomes. It does not complete the planned three-population grid. Retained traces were generated by the archived 11 September source; subsequent audit repairs have targeted regressions and do not retroactively change that source identity. Independently written verification scripts recompute all final and intermediate chart metrics, occupancy targets, staff constraints, release statuses and optimality claims, and check privacy implications with a separate CP formulation.

```
python -m pip install -r requirements-paper.txt
python -m pytest -q
python paper/assemble_evidence.py --retained-source
python paper/assemble_evidence.py --retained-source --no-site --partial \
  --inputs results/revised_evaluation_a.json results/revised_evaluation_b.
  json \
  --out results/revised_evaluation.json
python paper/make_tables.py
python paper/make_evaluation.py --manifest results/revised_evaluation.json
  --retained-source
python paper/make_figures.py --retained-source
cd article
latexmk -pdf -interaction=nonstopmode article.tex
```

7 Scope of conclusions

The implementation validates every released chart, keeps validated fallbacks, and supplies sound certificates for examined coercive cores under both the relaxation and the full model. It does not establish complete anti-capture screening, universal feasibility, optimal full-year schedules, empirically calibrated network behavior, fairness, privacy preservation, or improved wellbeing. The omission star permits $3 + 3$, $4 + 2$, and 6 ; its bound of three does not force two triples. A real pilot (see `docs/policy/pilot_evaluation_plan.md`), author declarations, and journal acceptance require evidence or decisions outside these synthetic experiments.